

Interacting Particles

Conor Brolly
40400295

In this report we produce a simulation of a system of interacting particles in a lattice using the Metropolis Monte-Carlo method. We produce an estimate of the probability for spins in the lattice.

1 System

The model we investigate is made of 64 particles in a square lattice of side length $L = 8$, each with a spin of $+1$ or -1 with coordinate s_i . Each spin interacts with every other spin.

The Hamiltonian of this system is given:

$$H(s) = -H \sum_{i=1}^{L^2} s_i - J \sum_{\langle i,j \rangle} s_i s_j \quad (1)$$

such that the sum over the pair $\langle i,j \rangle$ is over every pair of spins. Here, H is the magnetic field strength and J is the interaction energy.

We investigate the average magnetisation per spin, which we calculate as:

$$\overline{M} = \frac{1}{L^2} \sum s_i \quad (2)$$

2 Monte-Carlo Simulation

To estimate the average magnetisation per spin $\overline{M}$ for a given temperature, field strength and interaction energy, we produce configurations of spins with the Metropolis algorithm.

Trial moves are generated by either flipping all spins (with probability $\frac{1}{L^2+1}$) or flipping the spin of a single particle, chosen randomly.

We plot $\overline{M}$ against simulation step for the entire simulation and for the first 1000 steps to ensure that the model reaches a stable equilibrium and to ensure we can accurately discard data from before the equilibrium is reached.

As shown below, figure 1 shows us that the system reaches a constant equilibrium over the course of our simulation.

Figure 2 shows that the system reaches an equilibrium from around 100 simulation steps. In further simulations, we will ignore the first 200 timesteps to avoid the short initial period affecting our results.

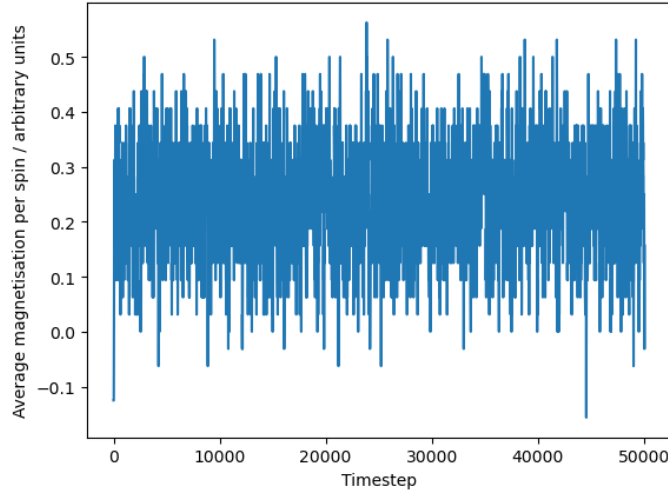


Figure 1: The average magnetisation per spin of 64 interacting particles in a square lattice with field strength $H = 4$, interaction energy $J = 8$ and a temperature of $t = 10$ in a Metropolis Monte-Carlo simulation of 50 000 timesteps.

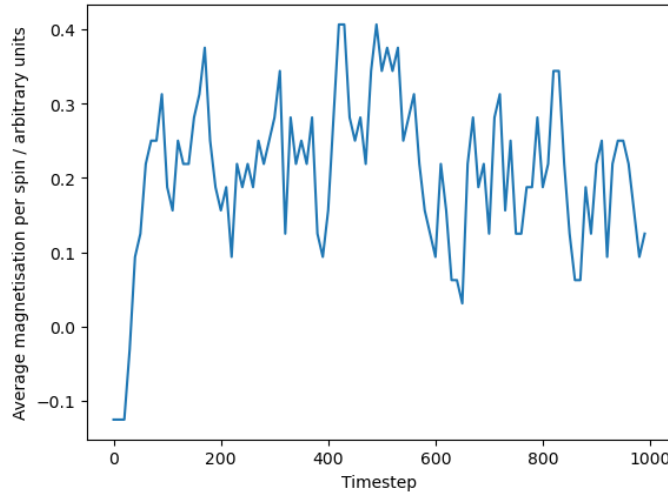


Figure 2: The average magnetisation per spin of 64 interacting particles in a square lattice with field strength $H = 4$, interaction energy $J = 8$ and a temperature of $t = 10$ in the first 1000 timesteps of a Monte-Carlo simulation. We can see that the magnetisation oscillates about a stable equilibrium for the duration of the simulation.

3 Estimating Probability

We can use our simulation data to estimate the probability density of a spin in the lattice by plotting a histogram of the average spins from our simulation, as seen in figure 3.

To estimate a probability density from this, we normalise the data so the total probability is 1 and plot a continuous line across the data. Figure 4 shows this distribution. It appears from our simulation that spins are distributed normally, although more investigation would be necessary to confirm this.

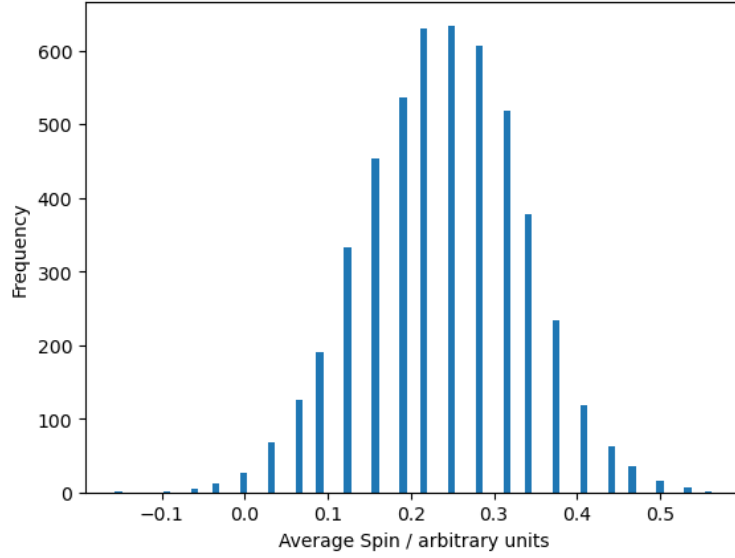


Figure 3: The average magnetisation per spin of particles in our system. The bar chart is plotted using 5000 data points from a 50000 Monte-Carlo Simulation at temperature $t = 10$, magnetic field strength $H = 4$ and interaction energy $J = 8$. We can see that the points from our simulation take discrete values.

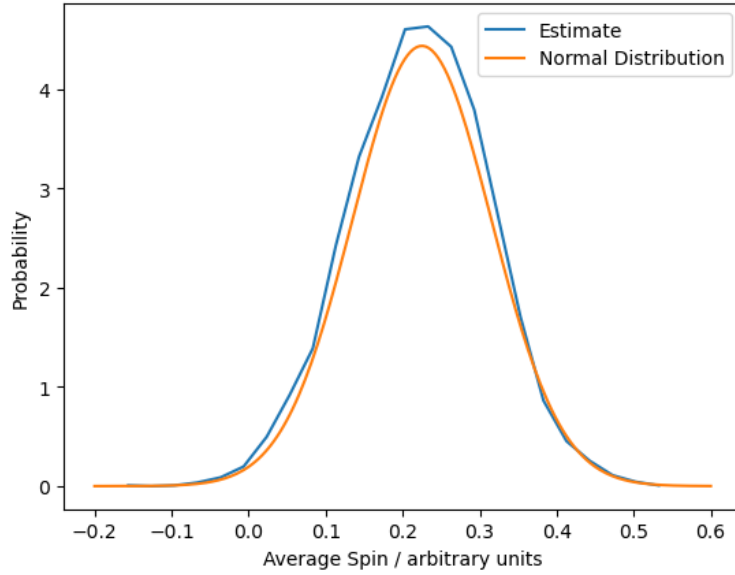


Figure 4: The estimate of the probability density for spins in our system at temperature $t = 10$, magnetic field strength $H = 4$ and interaction energy $J = 8$. This was estimated from a 50000-step Monte-Carlo Simulation. Also plotted on the graph is a normal distribution with mean 0.225 and standard deviation 0.09.